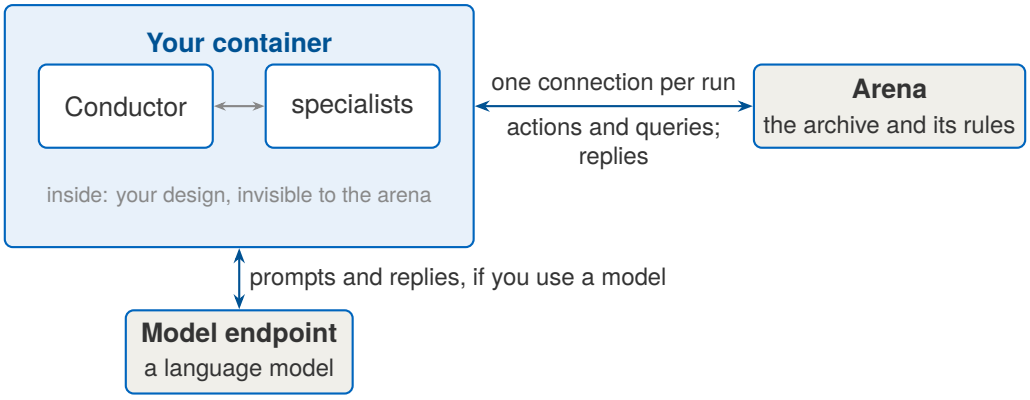
The arena at a glance

Who passes what to whom — enough for M1

IN2106 Multi-Agent AI Lab · version of 23 October 2026

The world's knowledge of music and computing has been sealed inside a vast digital archive: a maze of rooms full of volumes, with ambiguous titles, duplicates, outdated claims and evidence buried in long texts. Your Ensemble goes in as one team: each specialist takes on its own quest, and the Conductor holds the team together. What the archive says is what counts, even where the wider world disagrees.

The picture



- Your team ships **one container**. It is the only thing that talks to the arena, over **one connection per run**; a run is one instance of one quest.
- The arena talks to your team **as a whole**, never to one of your agents. How a message reaches the right specialist, and how your agents share what they know, is your design — that is what the Ensemble part of M1 describes.
- **Model calls never pass through the arena.** Your code calls a model endpoint whose address it reads from the environment: while you develop, Morpheus, with your CIT account and your own key; in graded runs, our endpoint, which counts the tokens.
- A graded run reaches only the arena and that endpoint, so everything your container needs must be inside the image.

One run

1. Your container connects to the arena.
2. The arena sends its **first message**: which quest this is, and what it asks of you.
3. Your container sends one message at a time and waits for the reply:
   - an **action** changes the world: moving, taking volumes, answering, skipping;
   - a **query** only reads, though the text you read is counted.
4. The reply to an action says what it did, and what you now see. The reply to a query carries what you asked for.
5. The arena's last reply says that the run is over; the arena then closes the connection.

Your team can also end a run itself, at any point, with an action of its own: whatever is still open then counts as skipped.

What a message from the arena carries

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{ quest:          which quest this is, and its goal
  outcome:        what your last action did
  trolley:        where it stands: the exits and their lengths,
                  in a room its volumes; what the trolley carries
  reading hall:   the volumes on its shelves
  setlist:        the questions, puzzles or claims still open, each
                  with its attempts left; its text comes in full only once
  spent:          steps, distance walked, text read
  done:           yes or no, and why }
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A sketch: the real messages have these parts, but their exact names and form may differ. Not every message has every part. The quest comes in the first message only, and what your last action did in every message after it. A quest for a single specialisation leaves out the parts it does not use; in the Overture, below, all of them come together. Each question, puzzle or claim comes in full only once, so your team has to keep the text. Model tokens are not in a message: our endpoint counts them.

Nothing in a message is addressed to one of your agents, and nothing lists which actions are allowed: the rules of each quest say that.

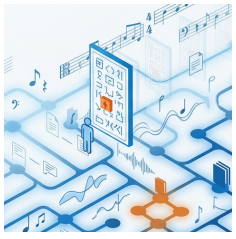
The four base quests



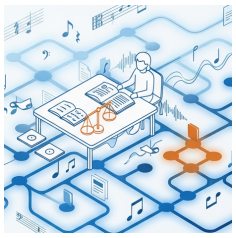
Night Round (the Improviser's quest, Navigation). Your agent explores rooms it has no map of. Every exit says how long its corridor is, but not where it leads. The task is to collect every volume and bring it to the reading hall, the archive's entrance, a trolley load at a time. What counts is the distance walked, and the tokens used. If your team ends the run before every volume is in the reading hall, only the volumes delivered count.



Reading Desk (the Musicologist's quest, Research). Your agent sits in the reading hall and is given questions about the volumes there. It answers each with a short fact and names the volumes that support the answer. What counts is whether the answer is right, whether the named volumes support it, how much your agent had to read to get there, and the tokens used.



Gate (the Player's quest, Strategy). Your agent stands at a locked door that sets it puzzles one after another, each described in a paragraph. It has one attempt at each puzzle and may skip any of them. What counts is how many puzzles it solves, and the tokens used.



Variorum (the Critic's quest, Investigation; teams of four). Your agent decides what to believe, and why. The volumes in the reading hall describe a music scene, and several sources cover the same albums. At the start your agent is given a set of claims, each with the passages that bear on it and the volume each passage comes from. For each claim it answers true or false, or skips it. What counts is +1 for a right answer, 0 for a skip and −2 for a wrong one, and the tokens used.

What passes each way:

Quest	The arena sends	Your container sends
Night Round Improviser	the exits where the trolley stands and their lengths; in a room, its volumes; what the trolley carries; the volumes in the reading hall	moves; volumes to take
Reading Desk Musicologist	the questions; the volumes in the reading hall; the text you ask for	queries; an answer to each question, with the volumes it rests on, or a skip
Gate Player	puzzles, one after another, each described in a paragraph	an answer to each puzzle, or a skip
Variorum Critic, teams of four	all the claims at the start, each with the passages that bear on it and the volume each passage is from	queries; true or false for each claim, or a skip

The Overture: your whole Ensemble

In the Overture your whole Ensemble plays one quest together, before M2; the Symphony, its full version, follows later. The run starts in the reading hall, where your team is given the questions. There is no map: every exit carries a sign naming the topic of the room beyond it, and the answers are somewhere in the archive. So your team explores with the trolley, looks at the titles and abstracts in the rooms it visits, brings the promising volumes back to the reading hall, the only place where volumes can be read in full, and answers each question with the volumes it rests on. Some corridors end at a locked door with a puzzle on it; solving it opens a shortcut. The run ends when every question has been answered or skipped. One message can carry everything in the sketch above: the trolley, the reading hall, and questions and puzzles on the same setlist.

In a test of one specialisation

At M2 and M3 each specialisation is also tested alone, on its own quest. The arena's first message always names the quest, so your container knows from the start whether it plays one specialisation's quest or the Overture, and can hand a single specialisation's run straight to that part.

For M1

If a diagram helps your Ensemble part, draw your container as the box above and fill it in: your parts, who passes what to whom, what is shared and which part holds the one connection to the arena.